

CHAPTER 1 RESULTS

1.1 Cohort Characteristics

TCGA-LUSC datasets for five modalities were retrieved (described in Section ??).

Gene expression and Copy number variation (CNV) retained the full labelled 178-sample cohort, whereas Deoxyribonucleic acid (DNA) methylation and miRNA expression had smaller assay overlap (Table 1.1). Across modalities, the classical subtype was the most frequent class and primitive the least frequent, motivating the use of macro-averaged classification metrics in subsequent analyses.

Table 1.1: Summary of omics modalities used in this study. Variation in sample size reflects assay availability and label overlap.

Modality	Configuration	Classical	Basal	Secretory	Primitive	Total
Gene expression	HiSeqV2, RNASeqV2 RSEM-normalised	65	43	43	27	178
DNA methylation	HumanMethylation450k beta values	29	14	22	11	76
miRNA expression	miRNA sequencing, log2(RPM + 1)	44	29	20	14	107
RPPA	Normalised protein abundance	38	28	28	18	112
CNV	GISTIC2 all-thresholded calls	65	43	43	27	178

1.2 Preprocessing Summary

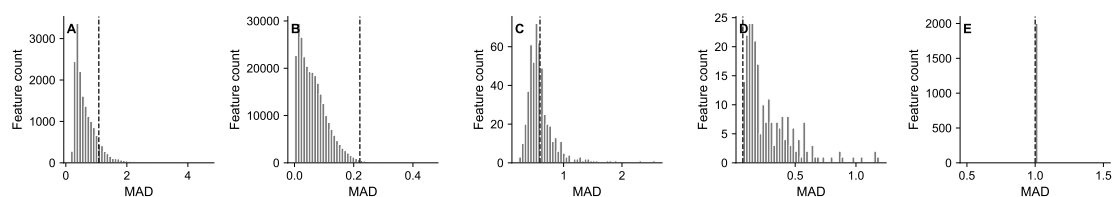


Figure 1.1: Feature variability distribution across processed omics modalities. Panels show MAD distributions for (A) gene expression, (B) DNA methylation, (C) micro RNA (miRNA) expression, (D) RPPA, and (E) CNV. Dashed vertical lines denote retained-feature threshold used for the processed matrix; for RPPA, all 238 antibodies were retained and the line denotes the minimum retained MAD.

Table 1.2: Preprocessing and feature-retention summary for each omics modality. Raw feature counts correspond to the input assay feature spaces. The MAD threshold is the score of the last retained feature in the processed matrix; for RPPA, where all 238 antibodies were retained, this value represents the minimum observed MAD among retained antibodies.

Modality	Raw features	Retained features	MAD threshold
Gene expression	60,660	2,000	1.0808
DNA methylation	485,577	2,000	0.2208
miRNA expression	1,881	200	0.5954
RPPA	238	238	0.0692
CNV	24,776	2,000	1.0000

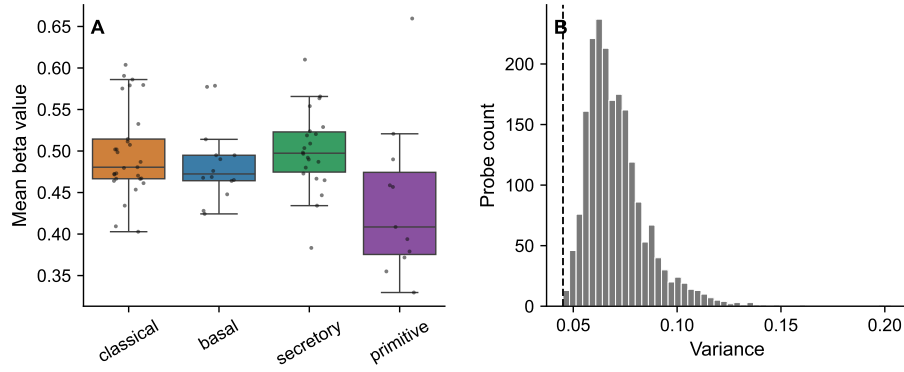


Figure 1.2: DNA methylation overview for labelled samples. (A) Mean beta value per sample stratified by subtype. Boxplots show median and interquartile range; points show individual samples. (B) Variance distribution across retained methylation probes. Dashed line denotes the minimum variance among retained probes in the processed matrix.

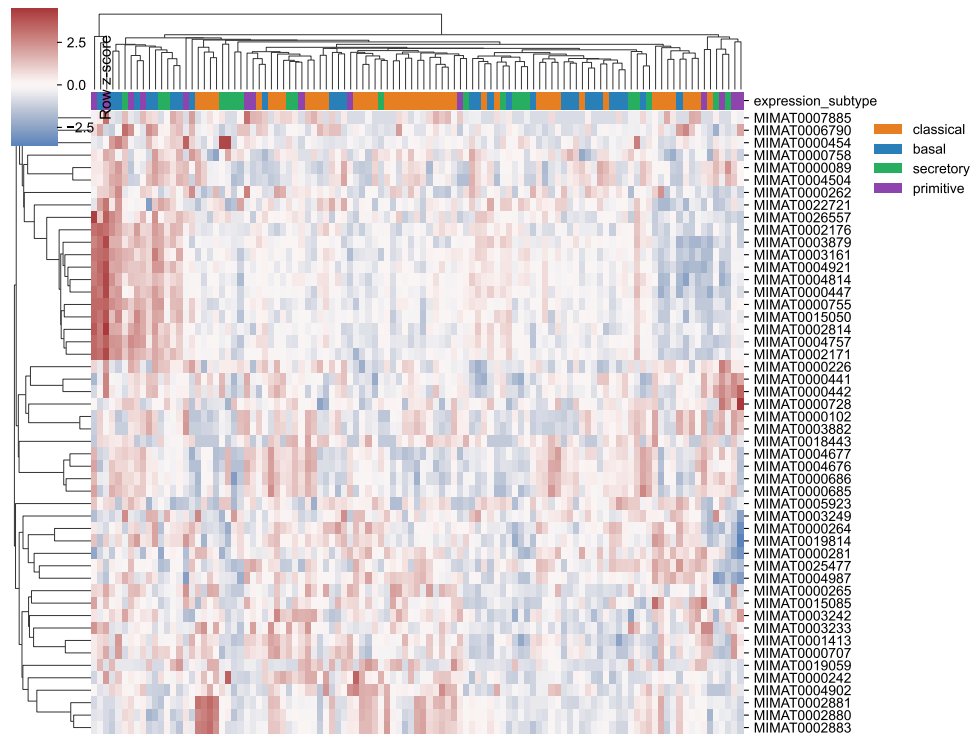


Figure 1.3: Clustered heatmap of 50 most variable miRNA features across labelled The Cancer Genome Atlas (TCGA)-Lung squamous cell carcinoma (LUSC) samples. Rows are miRNA features and columns are samples; values are row-scaled z-scores.

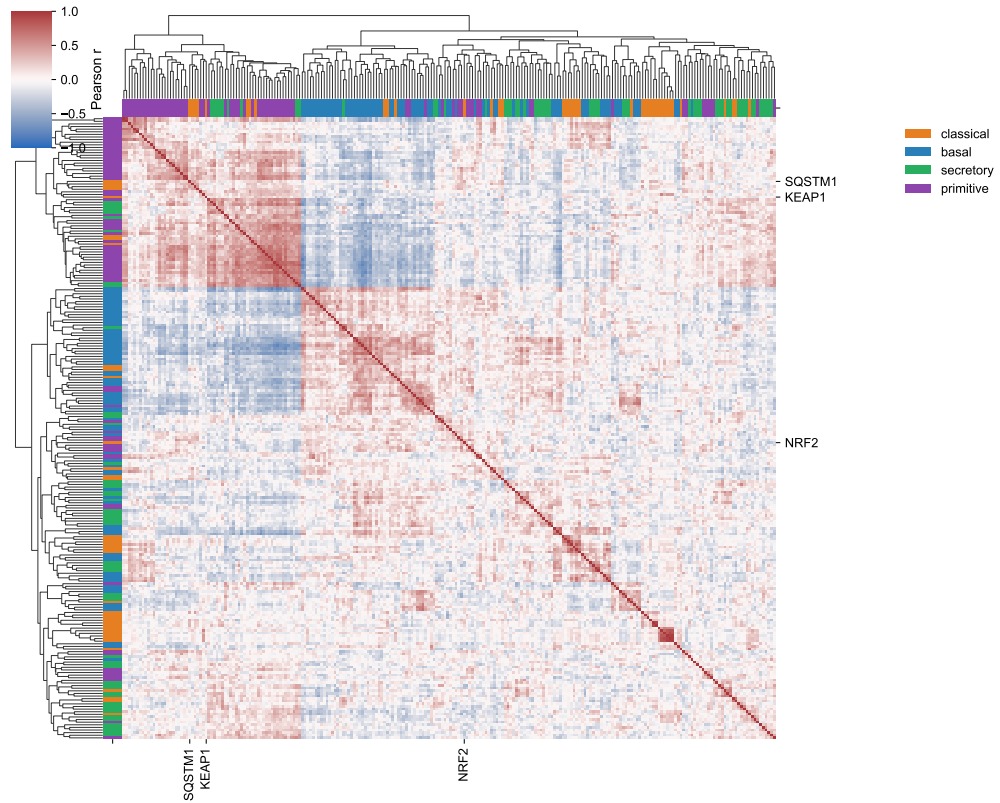


Figure 1.4: Hierarchically-clustered correlation heatmap across all 238 RPPA antibodies. Rows and columns represent antibodies, and colour denotes antibody-antibody correlation across RPPA samples. The top and side annotation strips indicate the expression subtype in which each antibody had the highest mean abundance.

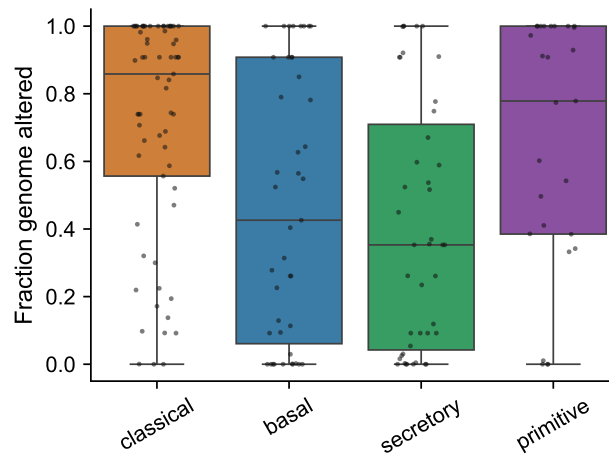


Figure 1.5: Fraction of retained CNV features with non-zero GISTIC2-thresholded copy-number state per sample, stratified by subtype. Boxplots show median and interquartile range; points show individual samples. Because the calculation is based on the 2,000 retained CNV features, it represents relative alteration burden within the processed CNV matrix.

1.3 Characterization of LUSC Subtypes

1.3.1 Clinical Characteristics

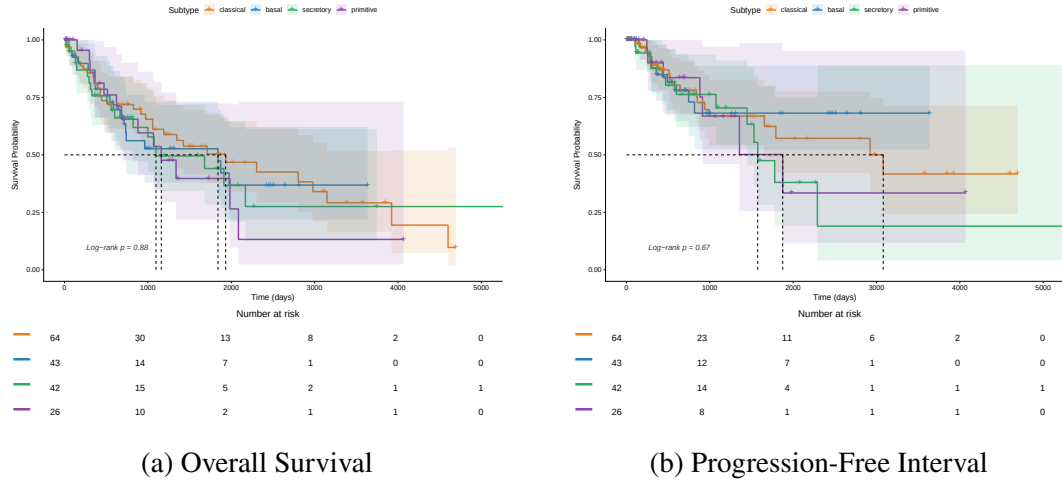


Figure 1.6: Survival curves showing Overall Survival (a) and Progression-Free Interval (b) for TCGA-LUSC samples (n=178) stratified by molecular subtypes. Median survival and 95% confidence intervals were computed by the Brookmeyer-Crowley method. Shaded regions denote 95% confidence intervals and risk table shows at-risk patient counts at selected time points.

Secretory and primitive subtype show the shortest median OS of 36 months and 38.1 months respectively, consistent with known proliferative and poorly differentiated biology of primitive subtypes (Hammerman et al., 2012; Wilkerson et al., 2010).

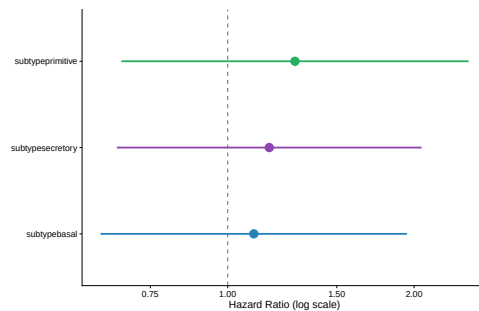


Figure 1.7: Hazard ratios for overall survival by LUSC molecular subtype from Cox proportional hazards regression, with classical subtype as reference (n=AVOCADO). Points denote hazard ratios and horizontal bars represent 95% confidence intervals. HR > 1 indicates worse prognosis relative to classical subtype. The proportional hazards assumption was verified using scaled Schoenfeld residuals (cox.zph global p=AVOCADO). Concordance index=AVOCADO

1.3.2 Immune Microenvironment

Mean EPIC cell fractions across LUSC expression subtypes

Top annotation shows BH-adjusted Kruskal-Wallis significance: * $q < 0.05$, ** $q < 0.01$, *** $q < 0.001$

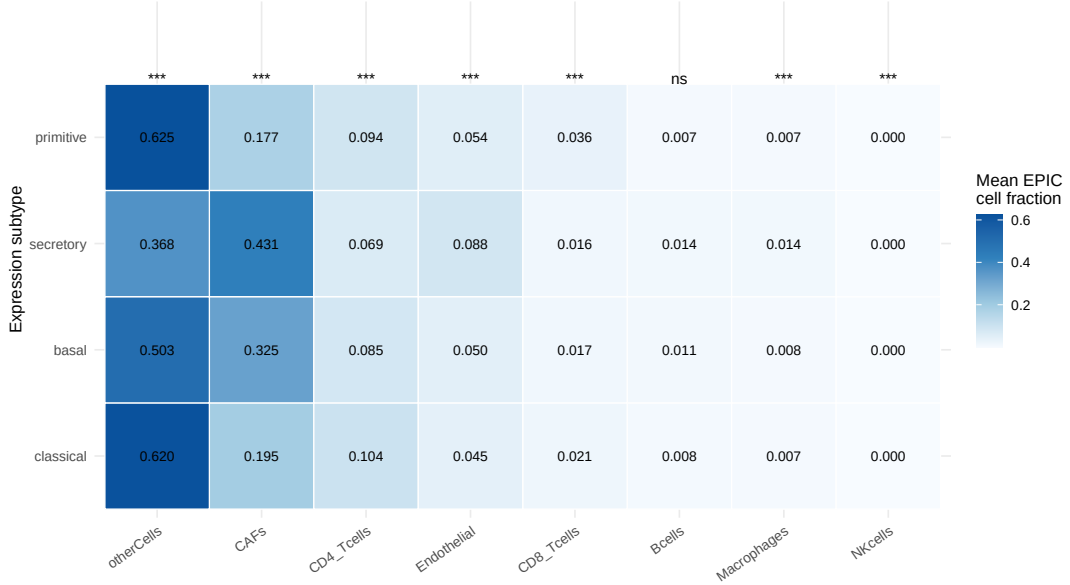


Figure 1.8: Mean EPIC cell fractions across LUSC subtypes ($n = 178$). Rows represent expression subtypes; columns represent EPIC-inferred cell compartments ordered by mean fraction. Numeric values show the mean fraction per subtype-cell-type combination. Significance annotations denote BH-adjusted Kruskal-Wallis q -values across subtypes (* $q < 0.05$, ** $q < 0.01$, *** $q < 0.001$; ns: not significant).

Secretory subtype showed the highest EPIC fractions for macrophages and CAFs; primitive subtype showed the highest fraction of uncharacterised cells (otherCells), consistent with its highly proliferative phenotype. B cells did not differ significantly across subtypes.

EPIC immune cell fractions by LUSC expression subtype

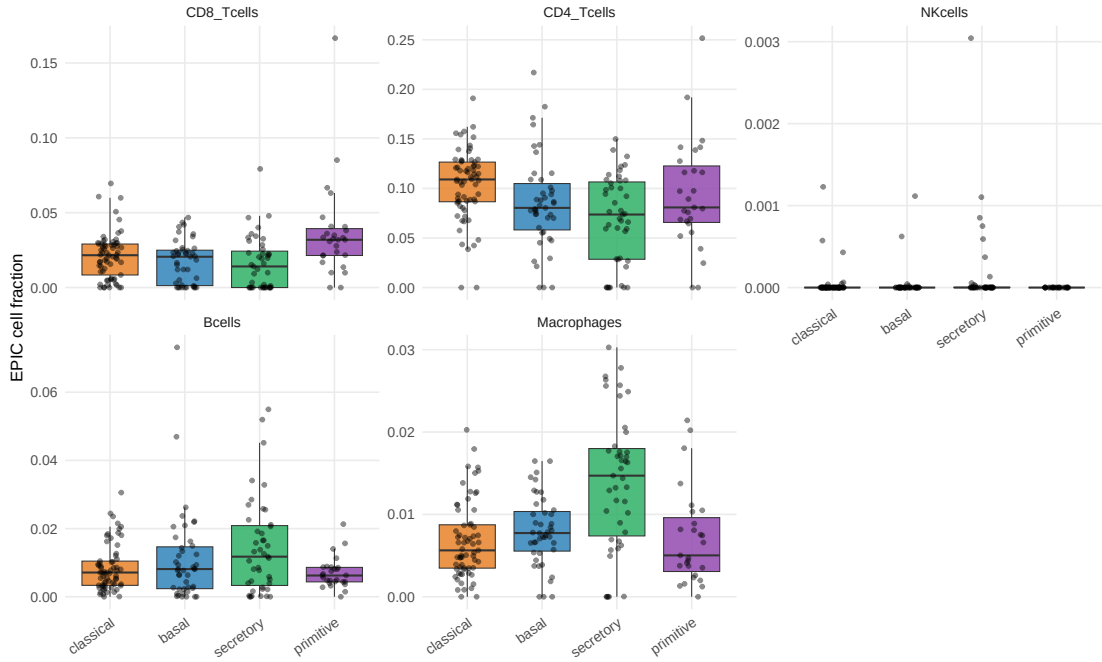


Figure 1.9: EPIC-estimated fractions for five cell populations across subtypes. Boxplots show median and interquartile range; individual points are samples.

xCell validation: immune enrichment scores by LUSC expression subtype

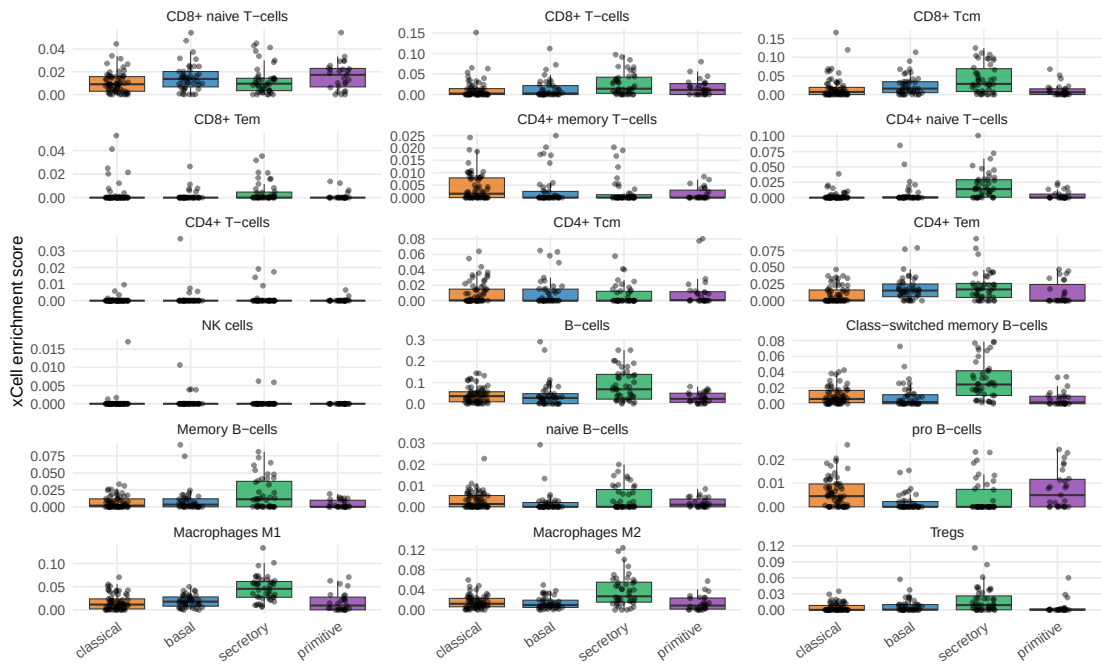


Figure 1.10: xCell enrichment scores for selected immune populations across subtypes. xCell scores are relative enrichment estimates, not cell fractions; scores across cell types are not directly comparable.

1.3.3 Somatic Mutation Landscape

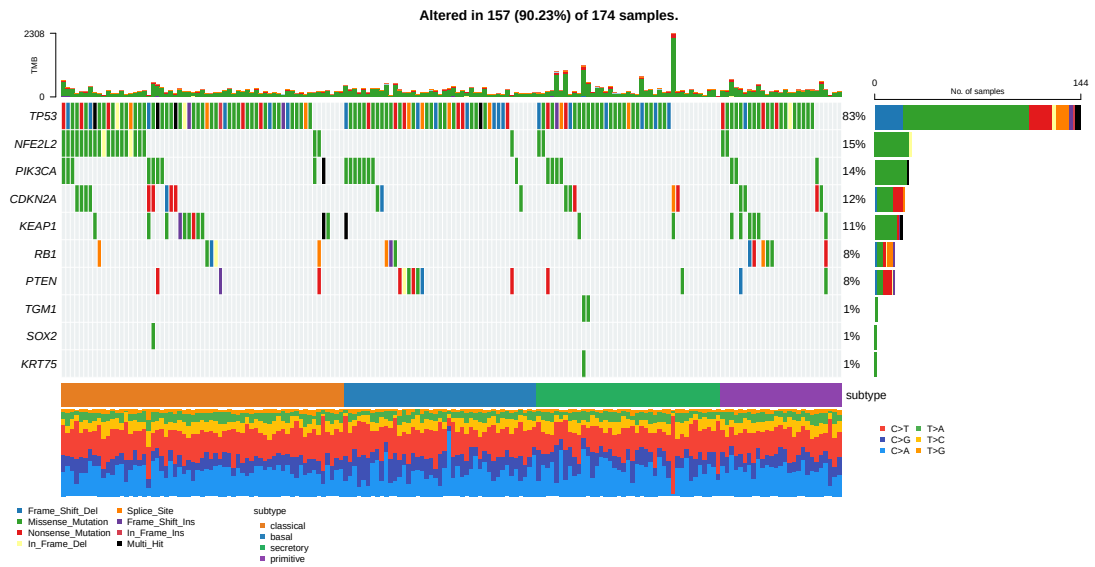


Figure 1.11: Oncoplot showing gene alterations across subtypes. Curated driver panel consisted of TP53, NFE2L2, KEAP1, RB1, PTEN, CDKN2A, PIK3CA, SOX2, FGFR1, TGM1 and KRT75.

TP53 was the most frequently mutated gene overall (Figure 1.11) but did not show subtype-specific enrichment, consistent with its role as a common subtype-independent driver alteration in LUSC.

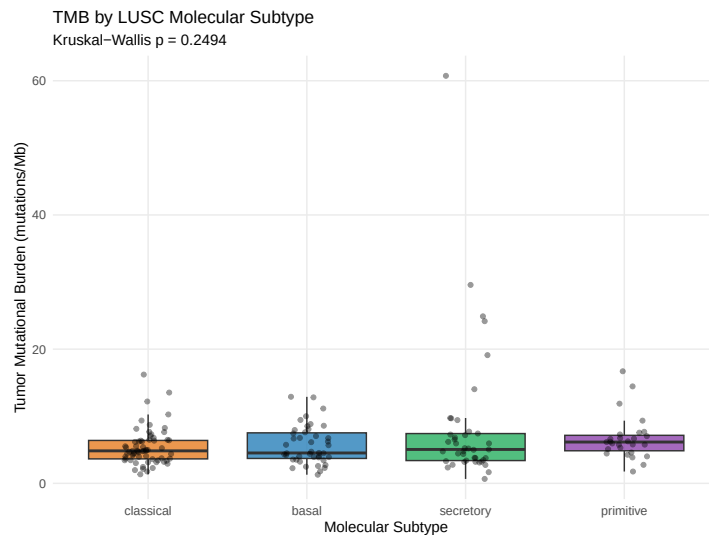


Figure 1.12: Tumour mutational burden by subtype.

TMB did not vary significantly across subtypes (Figure 1.12).

1.4 Dataset Separability

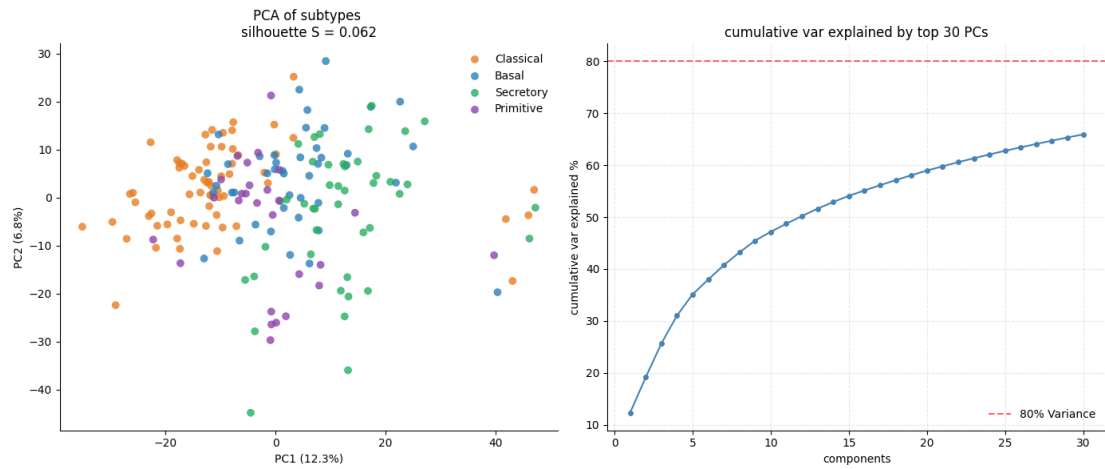


Figure 1.13: Each panel shows PC1 versus PC2 for one modality; variance explained by each component is shown on the axis label. Silhouette scores (S) computed on the top-30 principal components represent cluster separation independent of any supervised model.

1.5 Performance Across Omics Modalities

1.6 SHAP Feature Importance and Stability

1.7 Cross-Model Feature Overlap

1.8 Cross-Modality Feature Overlap

1.9 Pathway Enrichment

References

- Hammerman, P. S., Lawrence, M. S., Voet, D., Jing, R., Cibulskis, K., Sivachenko, A., Stojanov, P., McKenna, A., Lander, E. S., Gabriel, S., Getz, G., Sougnez, C., Imielinski, M., Helman, E., Hernandez, B., Pho, N. H., Meyerson, M., Chu, A., Chun, H.-J. E., ... group, D. w. (2012). Comprehensive genomic characterization of squamous cell lung cancers. *Nature*, 489(7417), 519–525. <https://doi.org/10.1038/nature11404>
- Wilkerson, M. D., Yin, X., Hoadley, K. A., Liu, Y., Hayward, M. C., Cabanski, C. R., Muldrew, K., Miller, C. R., Randell, S. H., Socinski, M. A., Parsons, A. M., Funkhouser, W. K., Lee, C. B., Roberts, P. J., Dang, T. P., Parejko, M. K., & Hayes, D. N. (2010). Lung squamous cell carcinoma mrna expression subtypes are reproducible, clinically important, and correspond to normal cell types. *Clinical Cancer Research*, 16(19), 4864–4875. <https://doi.org/10.1158/1078-0432.CCR-10-0199>